

APIs for Machine Learning

Lecture 1 - Introduction Notes

Day 1

1. What is an API?

API stands for **Application Programming Interface**. It is a set of rules that lets one piece of software (your app / your code) request a service or data from another piece of software (a server, library, or platform) — without needing to know how that other software works internally.

In simple words: an API is a **messenger**. You give it a request, it takes that request to the system that can fulfil it, and it brings the response back to you.

Real-life analogy: Restaurant

You (the customer) don't walk into the kitchen and cook your own food. You tell the **waiter** what you want, the waiter takes your order to the kitchen, and brings the finished plate back to you. You never see how the kitchen works — you just get the result. The waiter here is exactly like an API.



Fig 1: The API acts like a waiter between you (the user/app) and the server (the kitchen).

2. Why Do We Need APIs?

- **Reuse existing work** — you don't have to build a weather system or a payment system from scratch.
- **Save time & effort** — especially important in ML, where training a model can take days/weeks and huge datasets.
- **Hide complexity** — you just send data and get a result; you don't need to know the internal code, hardware (GPUs), or algorithm.
- **Allow different systems to talk to each other** — e.g. a Python script talking to a model built in a completely different language.
- **Scalability** — the heavy computation (like running a deep learning model) happens on powerful remote servers, not on your device.
- **Security** — the API only exposes what is safe to expose; your internal data/model stays protected.

3. Types of APIs

APIs can be grouped in two useful ways: **(a) by access level** and **(b) by communication style**.

(a) By Access Level

Type	Description	Example
Open / Public API	Anyone can use it, usually with an API key	OpenWeatherMap, OpenAI API
Internal / Private API	Used only inside one company/organization	An internal HR system API
Partner API	Shared only with specific business partners	Payment gateway partner API

(b) By Communication Style (most relevant for ML)

Type	How it Works	Used For
REST API	Uses simple HTTP requests (GET, POST, PUT, DELETE) and usually returns JSON	Most common; used by almost all ML model APIs (OpenAI, Hugging Face)
WebSocket API	Keeps a connection open so data streams continuously in real time	Live chatbots, streaming token-by-token text generation
GraphQL API	Client asks for exactly the fields it needs in one request	Apps needing flexible, precise data queries
gRPC API	Uses compact binary messages for very fast communication	High-performance ML serving between internal microservices

4. Where Are APIs Used? (Real-Life Examples)

Scenario	What Happens
Weather app on your phone	The app calls a Weather API to fetch live temperature/forecast instead of storing its own weather sensors
“Login with Google” button	The website uses Google's Auth API to verify who you are, without you creating a new password
Online payment (Easypaisa/JazzCash/Stripe)	Your app calls a Payment API to securely process the transaction
ChatGPT-style chatbot in an app	The app sends your text to the OpenAI API and displays the generated reply
Face unlock / photo tagging	The app sends the image to a Vision API (e.g. Google Vision, AWS Rekognition) which returns detected faces/objects
Spam filter in Gmail	A trained ML model is called through an internal API every time an email arrives

Practical Diagram: A Real Weather API Call

Below is exactly what happens, step by step, when a weather app on your phone shows you today's temperature:

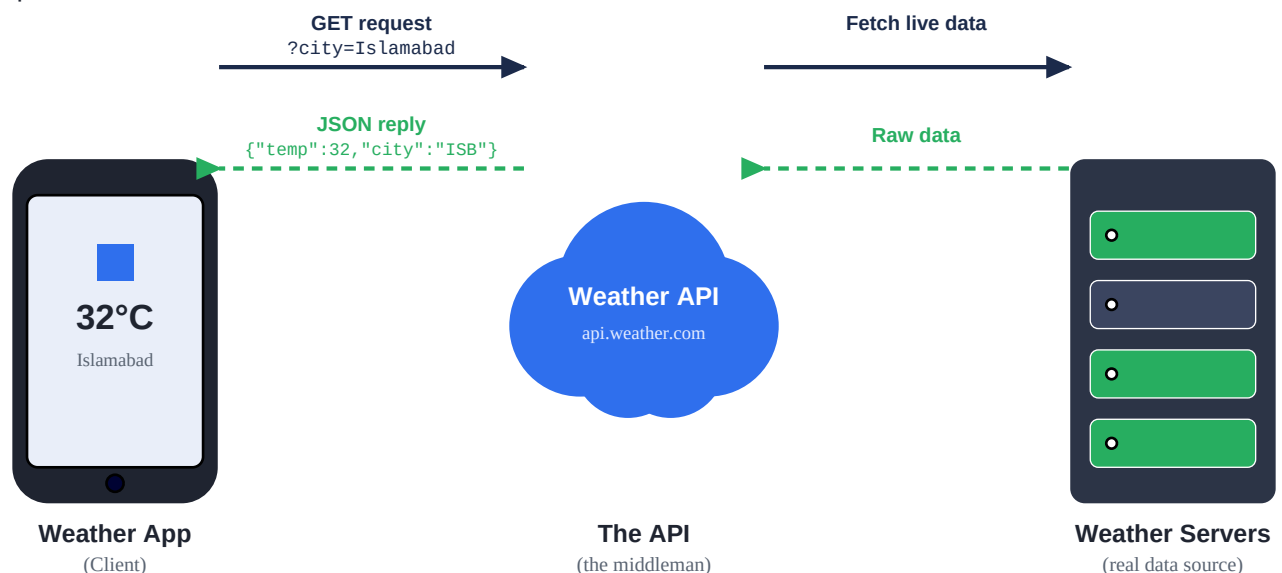


Fig 2: Client app sends a request with real parameters, the API forwards it to the actual weather servers, and a real JSON response travels back.

5. Where APIs Fit in Machine Learning

In ML, training a model needs huge data and powerful GPUs, which most apps/devices don't have. So instead of putting the model inside the app, the trained model is hosted on a server, and an **API is used to**

send data in and get predictions back. This is called **model serving** or **inference via API**.

Practical Example: Image Classification API

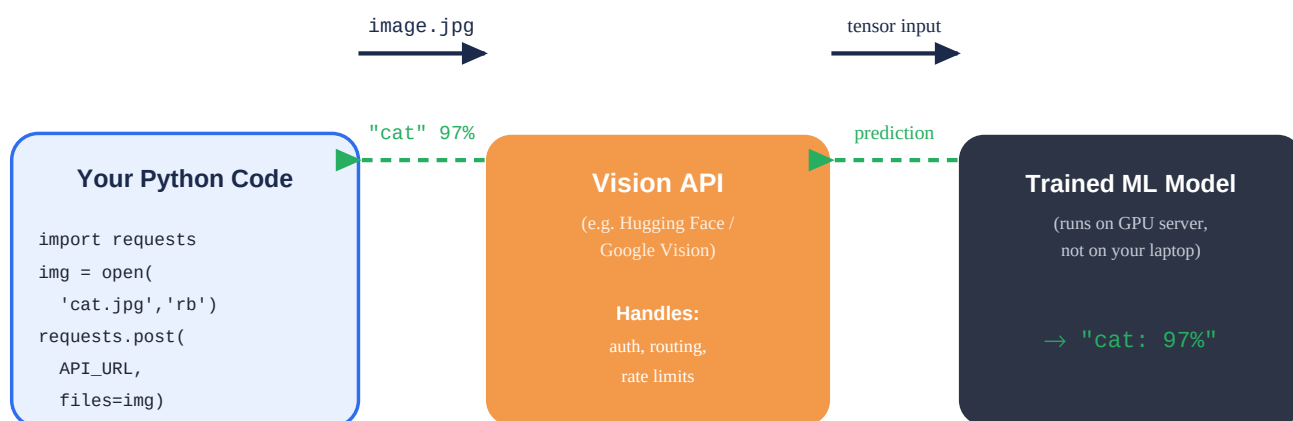


Fig 3: Your code never runs the model itself — it just sends the image over the API and receives the prediction.

Common Real ML APIs you will use in this course

API	Used For
OpenAI API	Text generation, chatbots, embeddings
Hugging Face Inference API	Thousands of ready-made ML models (NLP, vision, audio)
Google Cloud Vision API	Image labeling, face detection, OCR
AWS Rekognition	Object & face detection in images/videos

6. Key Takeaways

- An API is a messenger that lets two programs talk without exposing internal details.
- We use APIs to reuse work, save time, hide complexity, and connect different systems.
- The main types (by communication) are REST, WebSocket, GraphQL, and gRPC — REST is most common in ML.
- APIs are everywhere: weather apps, logins, payments, chatbots, and face recognition.
- In Machine Learning, APIs let you use powerful trained models (hosted on remote servers) directly from your own code, without training or hosting the model yourself.

Next class: How to actually call a REST API in Python using the requests library.